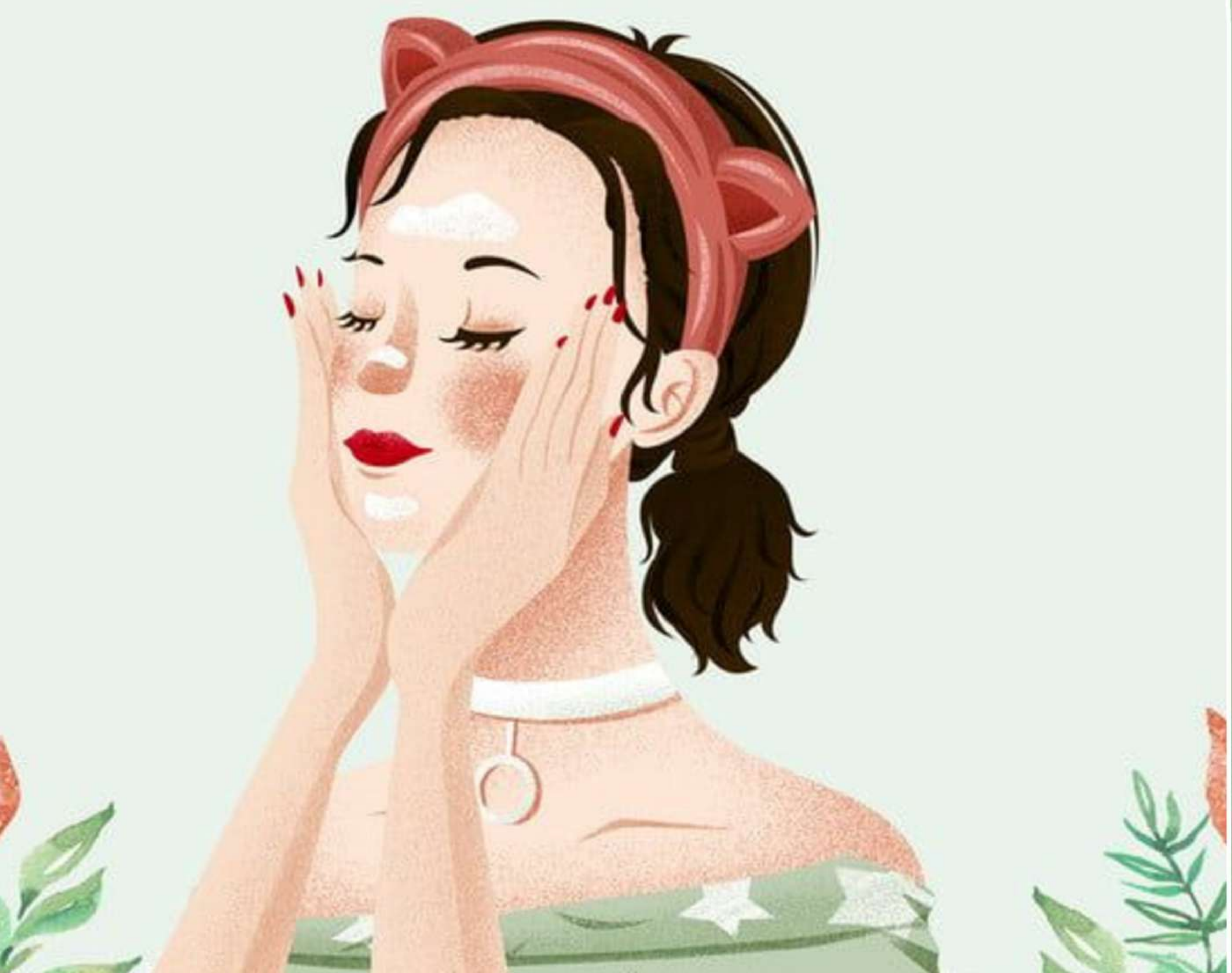


HABITS AND LIFESTYLE AFFECTING SKIN HEALTH

INTRODUCTION

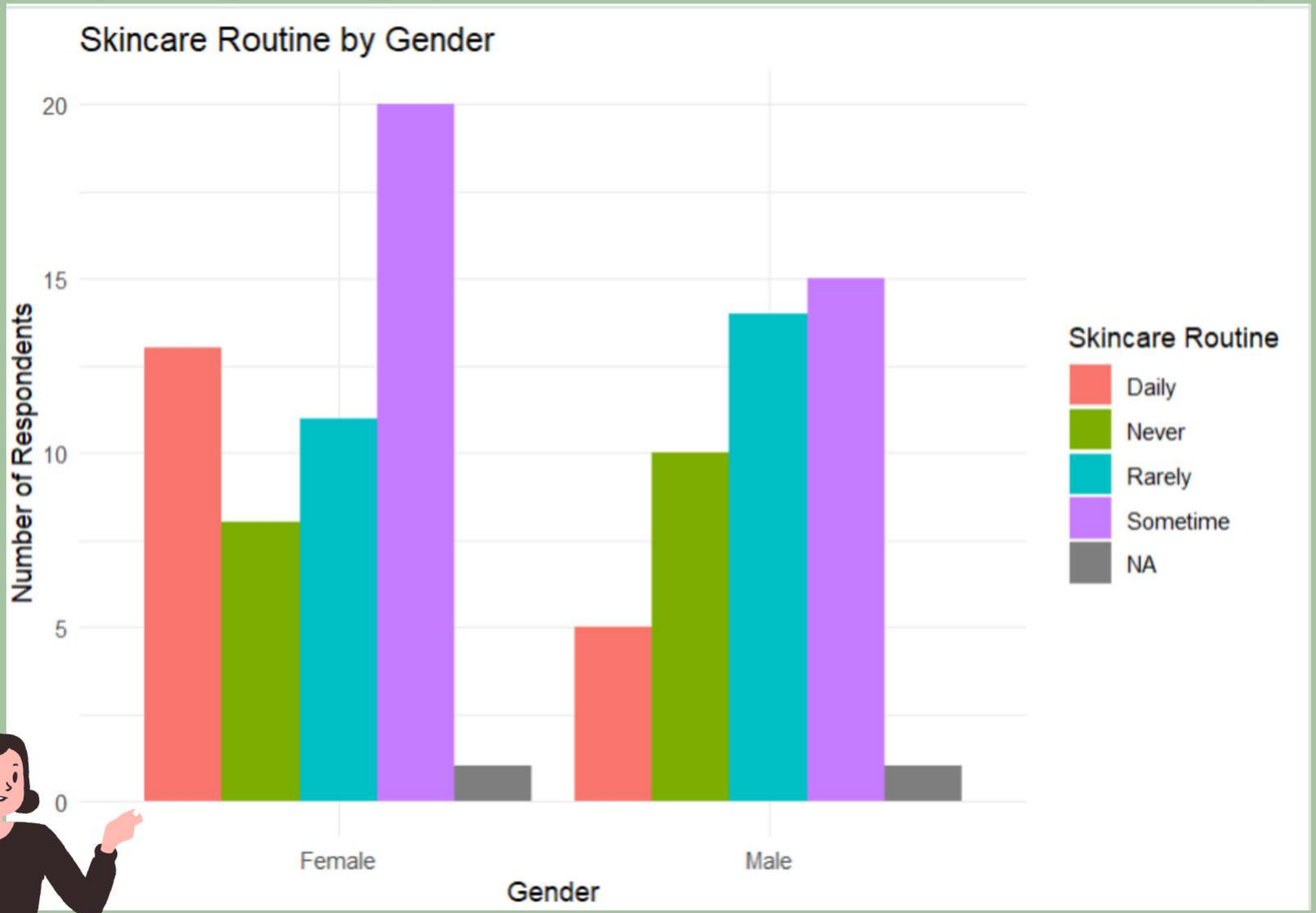
Today, skin health is getting worse day by day due to unhealthy lifestyle habits and increasing environmental challenges. Factors such as physical habits, environmental conditions, and irregular skincare routines all contribute to declining skin quality. This study focuses on analyzing the key lifestyle and environmental factors that influence skin health

OBJECTIVES :

1. Skincare Routine Distribution by Gender
2. Impact of Diet Habits on Skin Health
3. Effect of Lifestyle Factors on Skin Health
4. To study product usage and skin health
5. To analyze water intake effect on skin health
6. To study how different seasons affect skin condition.
7. To examine how lifestyle factors affect skin satisfaction.

SKINCARE ROUTINE DISTRIBUTION BY GENDER

- **Total respondents: 98**
- Female respondent=53
- Male respondent=45



- Females follow skincare routines more regularly than males
- “Sometimes” is the most common routine for both genders
- Males show higher irregular or rare skincare practices

IMPACT OF DIET HABITS ON SKIN HELATH:

Hypothesis:

H_0: Dietary habits (junk food and drink consumption) do not significantly affect mean skin health .

H_1 : Dietary habits significantly affect mean skin health .

TWO WAY ANOVA TEST:

```
> summary(anova_junk_drink4)
                       Df Sum Sq Mean Sq F value   Pr(>F)    
`junk food`             3  23.56   7.853   11.31 2.24e-06 ***
`drink consumption`     3  25.76   8.587   12.37 7.40e-07 ***
Residuals              91  63.18   0.694                     
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

DECISION : since p-value<0.05 for both factor ,reject null hypothesis(Ho)

CONCLUSION: Dietary habits significantly affect mean skin health .

EFFECT OF LIFESTYLE FACTORS ON SKIN HEALTH

Hypotheses:

H_0: Lifestyle factors (exercise, sleep, stress, and skincare routine) have no significant effect on skin health

H_1: Lifestyle factors have a significant effect on skin health.

TWO WAY ANOVA :

```
> summary(anova_lifestyles)
                    Df Sum Sq Mean Sq F value   Pr(>F)    
exercise             3  39.78  13.259  26.668 2.99e-12 ***
sleep                3   9.16   3.054   6.143 0.000786 ***
stress               3  17.49   5.831  11.729 1.67e-06 ***
`skincare routine`   3   3.80   1.268   2.551 0.04367  *  
Residuals           85  42.26   0.497                     
---
```

DECISION : since p-value<0.05 for all lifestle factor ,reject null hypothesis(Ho)

CONCLUSION: Lifestyle factors significantly affect skin health, with exercise having the greatest impact.

CORRELATION BETWEEN NUMBER OF SKIN PRODUCTS USAGE AND SKIN HEALTH

Hypotheses:

H_0: There is no significant correlation between skin product usage and skin health

H_1: There is a significant correlation between skin product usage and skin health

Test Used:

Spearman's Rank Correlation →

```
        Spearman's rank correlation rho

data:  data$`skin product` and data$`skin health`
S = 115803, p-value = 0.1934
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.1360819
```

DECISION : $p > 0.05 \rightarrow$ Fail to reject H_0

CONCLUSION: There is no significant correlation between skin product usage and skin health

EFFECT OF WATER INTAKE ON SKIN HEALTH (LINEAR REGRESSION)

Hypotheses:

H_0: Water intake has no effect on skin health ($\beta_1 = 0$)
H_1: Water intake significantly affects skin health ($\beta_1 \neq 0$)

Model: Skin Health = $\beta_0 + \beta_1$ (Water Intake)

Test Used:

Linear Regression

```
Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)        1.7962     0.3020   5.949 4.83e-08 ***
`water intake`     0.3914     0.1025   3.818 0.000244 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9769 on 92 degrees of freedom
Multiple R-squared:  0.1368,    Adjusted R-squared:  0.1274
F-statistic: 14.58 on 1 and 92 DF,  p-value: 0.0002439
```

DECISION : $p < 0.05 \rightarrow$ Statistically significant ,Reject H_0

CONCLUSION: Water intake has a significant effect on skin health

Skin Health = 1.7962 + 0.3914 .{Water Intake}

IMPACT OF SEASONAL VARIATION ON SKIN CONDITION

Hypotheses:

H_0: There is no association between season and skin condition changes.

H_1: There is an association between season and skin condition changes.

Test Used:

Pearson's Chi-square test of independence →

```
        Pearson's Chi-squared test

data:  skin_12
X-squared = 7.7194, df = 2, p-value = 0.02107
```

DECISION : $p < 0.05$ → There is association ,Reject H_0

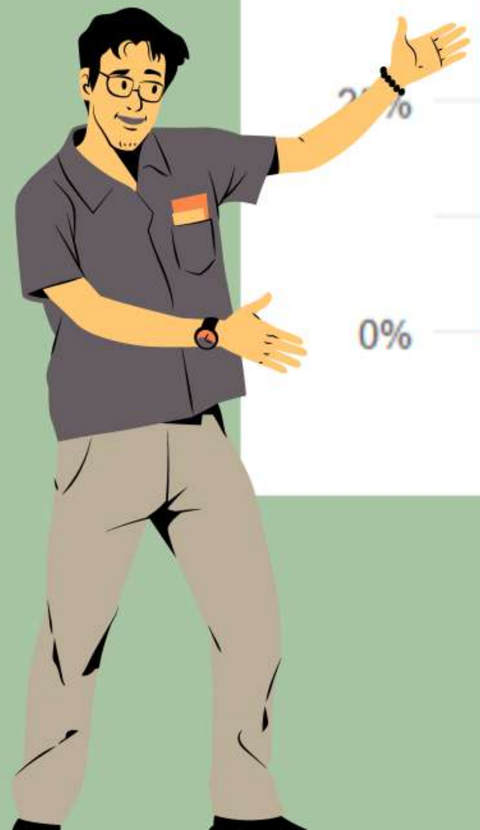
CONCLUSION: This indicates a statistically significant association between season and skin condition changes, showing that skin condition varies according to season.

LIFESTYLE INFLUENCE ON SKIN SATISFACTION

- People with poor skin satisfaction mostly have negative or neutral lifestyle effects.

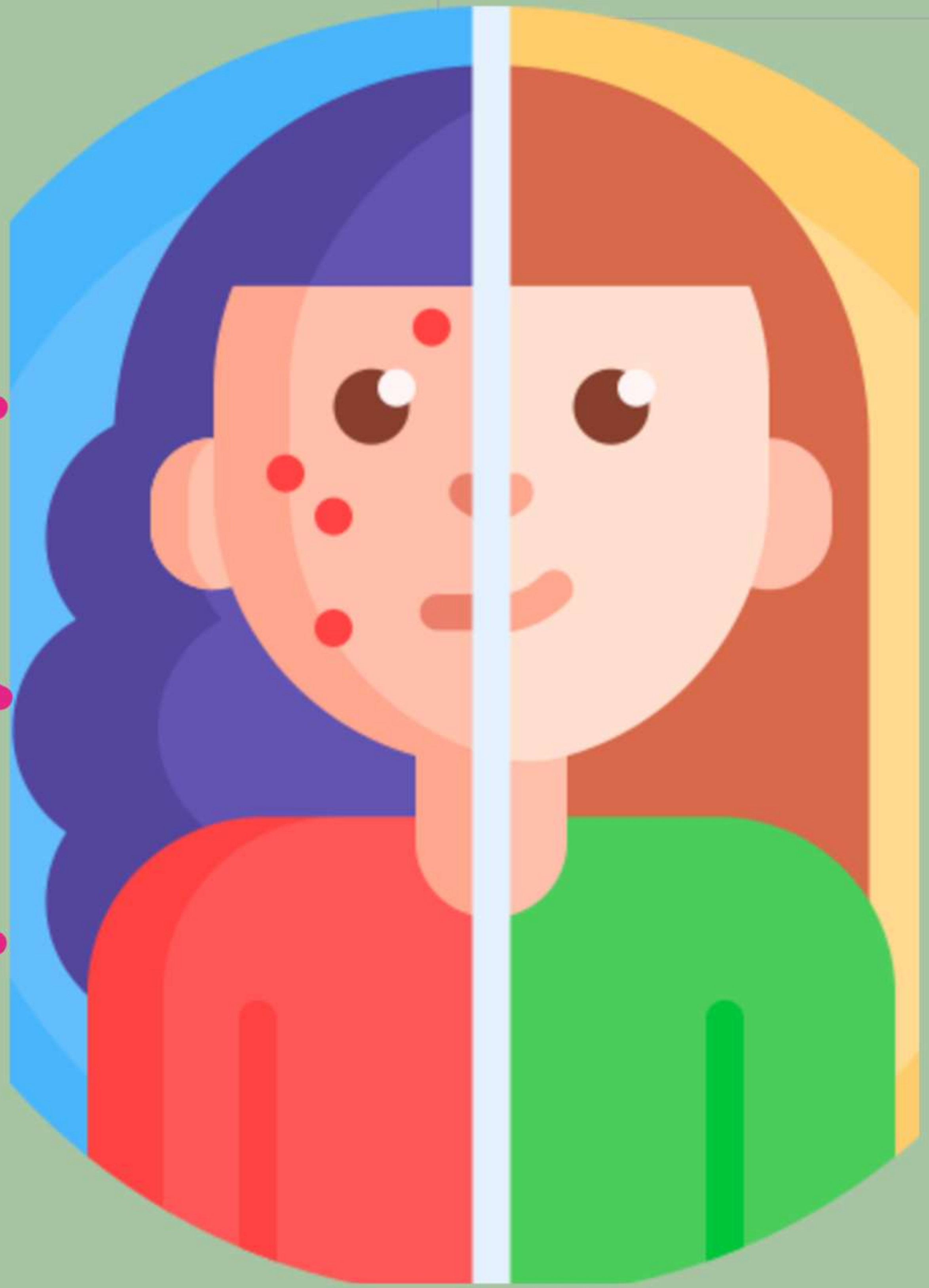
- People with good skin satisfaction show only positive lifestyle effects.

- This suggests that a healthy lifestyle is strongly linked to better skin satisfaction



Conclusion:

Unhealthy skin	Healthy skin
Physically inactive	Physically active
Poor diet	Healthy diet
Less water intake	Good water intake
Lack of sleep	Good sleep



THANK YOU

BORCELLE